

Challenge (Habanero)

- Q1.** A hiker travels 3.5 miles in 2.5 hours. What is the hiker's unit rate in miles per hour?
(A) 1.4 miles/hour
(B) 2.1 miles/hour
(C) 1.2 miles/hour
(D) 1.8 miles/hour
(E) 2.5 miles/hour
- Q2.** A car's fuel consumption is 25 miles per gallon. How many gallons will it consume to travel 375 miles?
(A) 10 gallons
(B) 12 gallons
(C) 15 gallons
(D) 18 gallons
(E) 20 gallons
- Q3.** The population of a deer species is directly proportional to the amount of available food. If 50 deer require 2000 pounds of food, how many deer can be supported with 4800 pounds of food?
(A) 80 deer
(B) 100 deer
(C) 120 deer
(D) 150 deer
(E) 200 deer
- Q4.** A group of friends want to share some trail mix equally. If they have 18 cups of trail mix and there are 6 friends, what is the unit rate of trail mix per friend?
(A) 2 cups/friend
(B) 3 cups/friend
(C) 4 cups/friend
(D) 5 cups/friend
(E) 6 cups/friend
- Q5.** A mountain climber ascends 2500 feet in 5 hours. What is the climber's unit rate in feet per hour?
(A) 400 feet/hour
(B) 500 feet/hour
(C) 600 feet/hour
(D) 700 feet/hour
(E) 800 feet/hour
- Q6.** A cyclist travels 24 miles in 2 hours. What is the cyclist's unit rate in miles per hour?
(A) 8 miles/hour
(B) 10 miles/hour
(C) 12 miles/hour
(D) 16 miles/hour
(E) 20 miles/hour
- Q7.** The cost of renting a camper van is directly proportional to the number of days rented. If a 7-day rental costs 945, what is the cost per day?
(A) 120 per day
(B) 135 per day
(C) 145 per day
(D) 155 per day
(E) 165 per day
- Q8.** A park ranger needs to patrol a trail that is 12 miles long. If the ranger's speed is 2.5 miles per hour, how many hours will it take to complete one round trip?
(A) 4 hours
(B) 4.8 hours
(C) 5 hours
(D) 6 hours
(E) 8 hours

Open Ended Questions (FRQ)

- Q9.** A group of hikers are planning a trip to a mountain peak. They will be hiking for 6 hours at a rate of 2 miles per hour. On the way back, they will be taking a shorter route that is 30% shorter than the original route. What is the total distance traveled by the hikers? Show your work and explain your reasoning.
- Q10.** A cyclist is riding a bike with a gear ratio of 3 : 1. If the cyclist pedals at a rate of 60 revolutions per minute, what is the speed of the wheels in revolutions per minute? Assume that the gear ratio remains constant and show your work.

Chef's Tips

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- Tip 1: Encourage students to read the questions carefully and identify the key concepts being tested.
- Tip 2: Remind students to use the correct units and to check their work for errors.
- Tip 3: Suggest that students use diagrams or visual aids to help them understand the problems.